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A fuzzy rank-based ensemble of CNN models for MRI segmentation

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ABSTRACT

Glioblastoma is the most common subtype of malignant tumors of the central nervous system. Segmentation of the brain tumor image is crucial for accelerating the diagnosis and treatment of a patient. In the paper, an advanced neural network ensemble based on a fuzzy ranking approach for tumor segmentation is presented using a combination of convolutional neural network (CNN) architectures, namely SegResNet, UNETR, and SwinUNETR. The proposed method uses fuzzy rank-based unification of deep learners by considering two nonlinear functions in decision-making, which helps to take into account the confidence in the predictions of the three base models. The proposed method is evaluated using the BRATS 2023 MRI dataset and outperforms the state-of-the-art methods, achieving an average Dice score of 0.885 ± 0.134. The statistical significance of the differences between the models and the ensemble is confirmed based on the Wilcoxon signed rank test, and the p-value is below 0.005.

1. Introduction

Cancer is a serious health problem. In 2020, cancer accounted for 18 percent of all deaths and is the second leading cause of death after heart disease. According to the European Cancer Information System [1], between 2010 and 2020, the number of common cancer cases in the population increased by 3.5% per year, partly due to an aging population. Brain tumors and other nervous systems are some of the main causes of cancer death among children and adolescents younger than 20 years of age [2].

Brain tumors are abnormal growths of cells that are thought to be multiplying without restraint. Slow-growing tumors are gradually identified and do not encroach on or penetrate adjacent tissues [3], yet can raise intracranial pressure and obstruct regular brain activity. In contrast, malignant tumors rapidly demonstrate and disrupt nearby tissue. According to the National Brain Tumor Society [4], glioblastomas make up more than 50% of all primary tumors of the central nervous system and are difficult to treat [5]. Survival rates and mortality statistics for glioblastoma have remained virtually unchanged for decades. Patients with low-grade gliomas may survive for months or even years, while those with high-grade gliomas usually have a much shorter life expectancy of weeks or months. The causes of glioblastoma are still unknown, but there are elements that can increase the likelihood of a brain tumor, such as radiation exposure and a family history of brain tumors.

Magnetic resonance imaging (MRI) is the gold standard for diagnosing patients with symptoms suggestive of a brain tumor. Compared to an alternative 3D imaging approach, namely Computed Tomography (CT), the MRI modality provides better visibility of soft tissue structures, allowing tumors to be located and measured during planning the appropriate treatment, and then evaluating the results of treatment [6]. Furthermore, MRI scanning is safer than CT because it does not use ionizing radiation from X-rays. MRI allows us to obtain much clearer images of the brain, since the result of MRI gives different types of contrasting images of tissues, such as T1-w, T2-w, FLAIR. With the help of this technology, three types of pathological tissues can be clearly distinguished in the image: the tumor nucleus (the most active cells), necrosis (dead cells inside the tumor nucleus), and edema (fluid accumulation).

Assessment of the total volume of pathological neoplasms and their exact localization in neurosurgery is crucial, and sometimes critical, to the choice of treatment tactics [7]. In modern radiomic approaches, determining the volume of interest (VOI) in a three-dimensional approach is the most important first step in the development of treatment planning. VOI determines the regions in which the radiological characteristics are calculated [8]. According to the experience of the dataset

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used [9], the manual segmentation of multimodal brain tumors taking about 60 min per subject and therefore this process is more expensive compared to computer-automated methods. The number of detected cases of brain cancer is increasing; therefore, more qualified specialists in the field of radiology are needed which sometimes are not available especially in the developing or underdeveloped nations.

As a result of the relevance of this topic, a number of studies on segmentation of brain tumors have been published in the past few years [6]. Machine learning based approach allows significantly increase the speed of processing medical images and highlight areas of interest. Segmentation is most often performed based on classical machine learning, but recently researchers have been predominantly using deep learning methods to increase the accuracy of segmentation.

The threshold-based segmentation method is a traditional, fast and easy-to-implement segmentation method [10]. Eman Abdel-Maksoud et al. [11] have developed a new approach to image segmentation, based on K mean, for images with MRI anomalies. Su Ruan et al. [12], with the use of support vectors, proposed a method to detect and localize brain tumor in MRI scans. However, these methods focus more on local details, such as image textures, which may not highlight features.

In recent years, deep learning methods have been used more often to analyze medical images [13]. One of the most used architecture of neural network in medical segmentation is the U-shaped models. Some popular models are DeepMedic [14], U-Net [15], V-Net [16], SegNet [17], etc. The U-shaped architecture learns well with fewer samples. Despite the small set of training data, it provides accurate segmentation results. This is confirmed by a large number of papers published on the segmentation process, in which U-Net is found to be an outstanding model [18–20]. As a result, this architecture is chosen as the basic segmentation structure [21]. It is worth mentioning that the applied research in the medical image analysis area meets the significant organizational problems related to regulatory, in particular, the main problem is inaccessibility of suitable for certification datasets, high cost, and what is most important low intensity of a data collection and labeling (especially in X-ray and CT modalities).

Next, Jyotismita Chaki and Marcin Wozniak [22] proposed the BTSCNet neural network for classifying brain MRI images using deep learning techniques. This model is based on a four-step approach that includes image preprocessing, feature extraction, classification, and postprocessing. BTSCNet uses a combination of convolutional layers to efficiently extract significant features from images, applying state-of-the-art activation functions to improve nonlinearity and overall performance. BTSCNet demonstrates high accuracy in classifying various types of brain tumors.

In [23] the authors present an enhanced instance segmentation method for automated analysis of brain tumor MRI scans, using a modified Mask Region-based Convolutional Neural Network (Mask R-CNN). This approach incorporates attention mechanisms and a Feature Pyramid Network (FPN) to effectively extract tumor characteristics. The architecture employs ResNet50 as the backbone, augmented with Squeeze-and-Excitation Networks and Concatenated Attention Networks to focus on critical image regions. The segmentation process involves generating region proposals through a Region Proposal Network (RPN) and applying RoIAlign for precise delineation of tumor morphology. The experimental results demonstrate the performance of the model, achieving an accuracy of 90.72%, a recall of 91.68%, and a mean intersection over union (IoU) of 94.56%, significantly improving the quality of MRI tumor segmentation.

In [24] the authors introduce an automatic brain tumor diagnosis system that uses a convolutional neural network (CNN) for the detection, classification, and segmentation of glioblastomas in MRI images. The system is structured in two stages: the first stage focuses on detecting and classifying brain MRI images into categories such as normal, high-grade glioma (glioblastoma), and low-grade glioma. This stage achieves a maximum accuracy of 99% using 1800 images from the BraTS 2017 dataset. The second stage is dedicated to tumor segmentation, utilizing skip connections and residual units to enhance performance. The study highlights the system's ability to integrate local and global features, improving the overall accuracy and efficiency of brain tumor diagnosis.

Wisal Zafar et al. [25] present an approach for diagnosing brain tumors using a hybrid deep learning model that combines convolutional neural networks (CNN) and recurrent neural networks (RNN). The results demonstrate that the hybrid model is superior to traditional methods and provides high accuracy in classification and segmentation tasks.

Based on the above discussion, the present paper aims to generalize an earlier proposed fuzzy rank-based ensemble method [26] for the image segmentation task. As a benchmark, the segmentation of the glioblastoma task has been chosen. In scope of the research, we keep in mind the limitation on the amount of data used for a neural network training and the training time to mimic limited resources in applied research. The proposed method, as mentioned, is based on a fusion of three state-of-the-art neural networks, namely SegResNet [27], UNETR [28], and SwinUNETR [29] models, using the fuzzy rank-based ensemble.

The paper is organized as follows. In the first section, we provide an introduction to the topic, where we discuss the background and significance of the research. After that in Section 2, we describe the methodology and material used in our study, including the research design, the description of the dataset, and the results analysis techniques. Following the methodology section, in Section 3, we present the results of our study and compare them with some recent approaches. Finally, in Section 4, we conclude the paper by summarizing the main findings and suggesting areas for future research.

2. Material and methods

2.1. Dataset

For the first time, a set of images of brain tumors with multimodal segmentation (BRATS – Brain Tumor Image Segmentation) was presented at the International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) in 2012 [9]. The latest version of the Brain MR Image Synthesis for Tumor Segmentation (BraSyn) dataset was introduced in 2023 [30]. BraSyn-2023 is based on the RSNA-ASNR-MICCAI BraTS 2021 image set [31]. This is a collection of preoperative glioma data consisting of multi-parametric (mpMRI) clinical scans from multiple institutions and various scanners using traditional brain tumor imaging protocols. Each case is provided as four processed MRI scans (T1, T1Gd, T2, and T2-Flair); the thickness of the image slice is 1–6 mm. Thus, the set collected provides real-world examples of MRI scans for the development of a reliable and automated method.

BraTS, together with internationally recognized neuroradiology experts, identified three tumor subregions (Fig. 1):

1. The enhancing tumor (ET): peritumoral edematous (infiltrated) tissue, a tumor site with an increased T1 signal in postcontrast images compared to precontrast images.
2. Nonenhancing tumor core (TC): includes all parts of the tumor core, necrosis.
3. Surrounding non-enhancing flair hyperintensity (or whole tumor - WT): "vasogenic edema" surrounding the tumor.

The tumor area of interest is small relative to the entire image and difficult to cover. The volume of the three tumor subregions in the test subset is shown in Table 1, where the imbalance among the various tumor subdomains is clearly visible. At the same time, glioma and especially its classes of tissues have complex boundaries and structures due to the special penetrating ability of malignant tissues to healthy ones, making it difficult to accurately determine its clear boundaries.

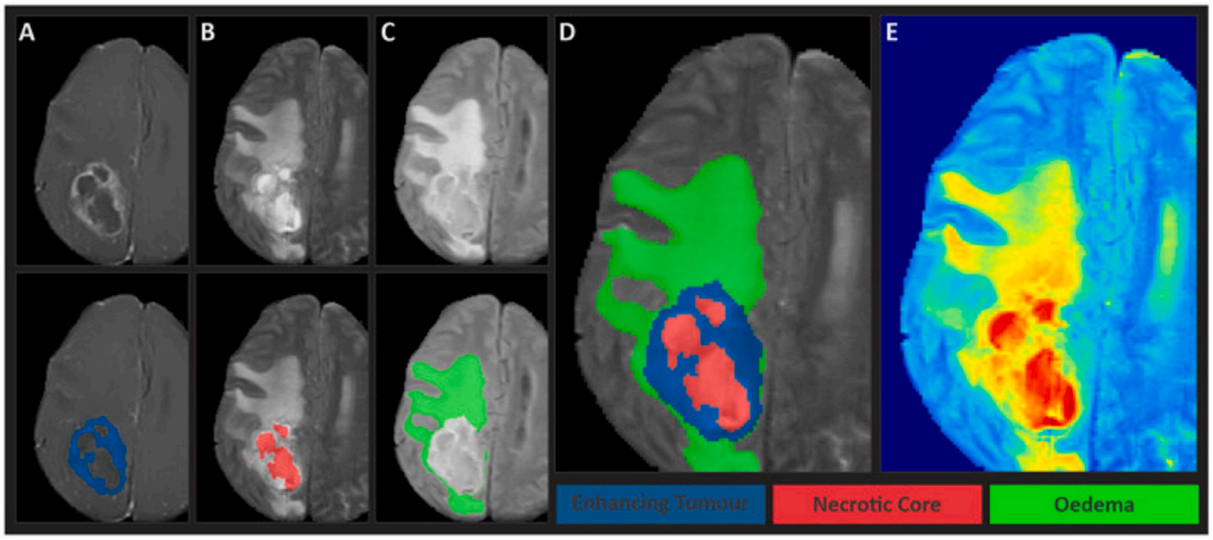

Fig. 1. Segmentation of pathological cells (image from [30]). Panel A: enhancing tumor (ET - yellow). Panel B: non-enhancing necrotic tumor core (NETC –red). Panel C: combined surrounding sub-region (SNET - blue). Panel D: three labels overlaid on MR image highlighted. Panel E: RBG color model.

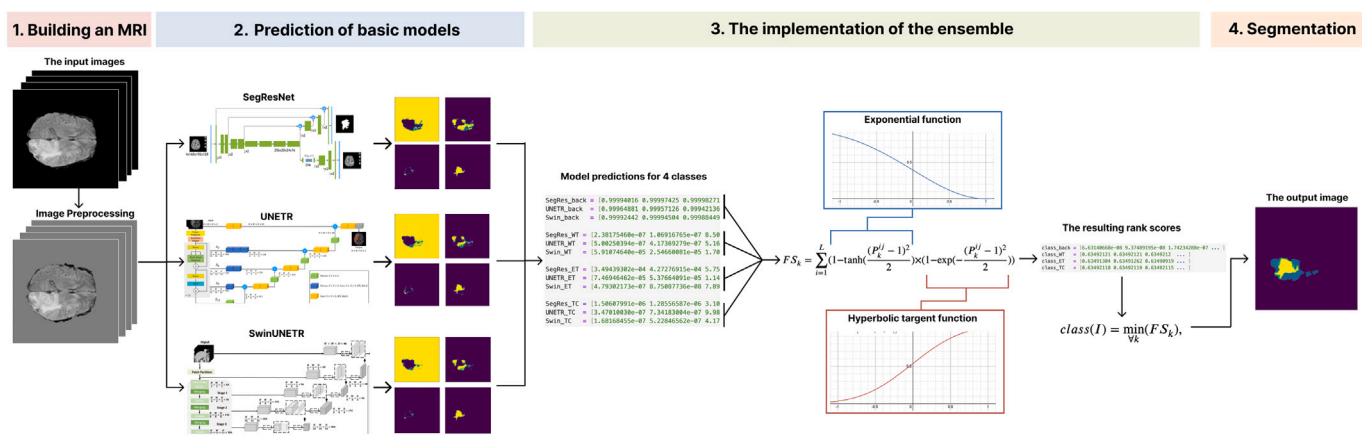

Fig. 2. The pipeline of the proposed method used for a glioma segmentation on an MRI scan.

Table 1
Statistic of volumes in the test subset.

	WT, cm^3	ET, cm^3	TC, cm^3
count	130	130	130
mean	69.8717	24.0307	10.0829
std	45.3459	17.2500	10.1371
min	3.0590	0.1570	0.0000
25%	35.3717	10.4763	1.9945
50%	60.9625	21.3880	7.2965
75%	103.1705	35.0147	15.9070
max	193.2260	74.5210	47.6340

As part of the study, to train selected neural network models, we introduce restrictions on the amount of data and training time in order to simulate limited resources in applied research. The training, validation and testing samples include 300, 30 and 130 images, respectively.

2.2. Proposed approach

The pipeline of the proposed approach in presented in Fig. 2. In the first stage, the MRI scans are pre-processed for four modalities. At the second stage, three basic models are introduced, namely SwinUNETR, UNETR and SegResNet. The output of the model looks like the probability of belonging to one of the four classes (the brighter the color, the closer to 1). In the third stage, the predictive estimates of the underlying neural networks (base models) are combined using an ensemble based on fuzzy ranking to obtain the final segmentation predictive score.

2.2.1. Base models

Three-dimensional CNNs have been successfully used in several medical image segmentation tasks and have recently become the main models for brain tumor segmentation due to its impressive high performance in solving classical computer vision problems, as well as in analyzing medical images compared to other models. U-shaped models are widely used in medical images segmentation. As a result, the following three models are selected.

1. SwinUNETR [32] is an image segmentation model combining the Swing Transformer architecture proposed by Liu et al. [33] and the U-Net architecture [15]. This model combines convolutional-transformer type blocks, which allows the model to efficiently transmit information at different levels of abstraction, and encoder and decoder mechanisms to extract features at different levels of resolution and effectively combine information.
2. Ali Khatami Zadeh and his colleagues decided to upgrade the architecture of the U-Net model and developed a new solution called UNEt TRansformers (UNETR) [28]. This solution is based on the use of transformer model, which allows to efficiently

process images and highlight objects on them. In addition, the authors of UNETR have proposed new methods for training this neural network, which helps to achieve better results on various image processing tasks.

3. The third model, SegResNet, is based on autoencoder regularization [27]. A feature of this architecture is that compared to previous models, an asymmetrically large encoder is used to extract deep image features and an additional branch, an auto-encoder, has been added. The auto-encoder adjusts the sampling rate of the input display of its objects at a lower resolution. This eliminates the need to study up-sampling. During the output, only the main part of the segmentation encoding-decoding is used.

Due to high memory consumption, the input shape for each model is configured as $[4 \times 96 \times 96 \times 96]$, where B is the batch size, 4 is the number of modalities and 96 is the patch size. However, the initial size of the MR images of the dataset used can be up to $240 \times 240 \times 155$. The final prediction during validation and testing is evaluated using the sliding window approach with an overlap ratio 0.5.

The model prediction has four projections (C, X, Y, Z), where C is four planes with the probabilities of segmentation of three classes of glioblastoma and background. The value of each voxel belongs to the interval [0,1], where 1 is the highest probability of belonging to a class. The resulting output data is adapted for further application of the ensemble.

2.2.2. Fuzzy rank-based ensemble

The model uses the concept of an ensemble of deep models based on fuzzy ranks as developed by Manna et al. [26]. The proposed ensemble method uses a fuzzy rank-based unification of deep learners by considering two nonlinear functions in decision estimates. Unlike the simple fusion schemes found in the literature, the proposed ensemble method makes final predictions for test samples, taking into account the reliability of the predictions of the base models. This paper presents an extension of the 2D image classification method to the task of segmentation of a 3D medical image. A prediction from three neural networks (i.e., base learners) is received at the input of the fuzzy ensemble.

The probability of a voxel classification to one of the classes can be written as following.

$$\sum_{k=1}^{C} P_k^{ij} = 1, \forall i = 1, 2, 3. \tag{1}$$

where C is the number of classes, i is the number of base models, j is the number of voxels.

The domain of definition of the functions will be [0, 1] as $P_k^{ij} \in [0, 1]$.

Two nonlinear functions were chosen to generate fuzzy ranks. An exponentially decreasing function takes into account the deviation from the goal for a class that has a prediction probability, when it reaches 1, the function goes to zero (Fig. 3a). The tangentially increasing function determines the reward for large indicators of the probability of belonging to the target class. The reward increases as x increases, eventually becoming 1 (Fig. 3b).

The product of the selected functions allows you to take into account rewards and deviations for a specific assessment of the reliability obtained using three models. The resulting rank scores will display the level of confidence in a particular class, as it is the result of fuzzy ranks generated by two different types of functions (Fig. 3c). Now, the fused score (FS) tuple is:

$$FS_k = \sum_{i=1}^{L} (1 - \tanh(X)) \times (1 - \exp(-X)),$$
$$X = \frac{(P_k^{ij} - 1)^2}{2}, \tag{2}$$

where $\forall k = 1, 2, \ldots, C$.

Then we find the class that has the lowest merged score and consider it the 'winner'.

$$class(I) = \min_{\forall k}(FS_k), \tag{3}$$

One of the four classes will be assigned a 'winner' to the pixel value. Thus, it will be possible to classify the voxel, i.e., segment the entire 3D MR images.

It is worth noting that any particular or specific weight to each model or to each pixel in the image is not assigned. Inside the model, the weight is set dynamically for estimating a pixel belonging to a particular class.

2.3. Model training pipeline

2.3.1. Preprocessing

In order to expand the dataset and improve the learning ability of the model, some image preprocessing methods are used in the work. The transformations used are described below:

1. Interval adjustment using pixdim = (1.0, 1.0, 1.0) based on an affine matrix.
2. Normalization of input data based on subtraction of the average value of the input image. This transformation can normalize only non-zero values.
3. Three-dimensional cropping of a random fragment of an image of dimensions [96, 96, 96], the image itself has a size [240, 240, 155].
4. Arbitrary flipping of the image on each axis with a probability 50%.
5. Arbitrary scaling of the intensity of the input image with a probability of 10% based on the formula $v = v * (1 + coefficient)$, where the coefficient is randomly selected in the range (−0.1, 0.1).
6. Arbitrary intensity offset with a randomly selected offset in the range (−0.1, 0.1).

The above transformations have been applied to the data to train the base models. For validation images, there are only two first points, as shown in Fig. 4

2.3.2. Training

The models are trained separately using the following methodology. For training, we have chosen AdamW [34]. The process of training each model within the framework of the work is limited to 150 epochs. Validation checks took place every 10 epochs, ran for only 20 batches, at the size of the batches [96, 96, 96]. Furthermore, according to the limitation of the computational platform used, the training time was limited to 10 h.

Adamw is a stochastic gradient descent method that estimates first- and second-order moments adaptively. The method also includes a technique for adjusting the attenuation weights, in accordance with established methods. The main goal of AdamW is to minimize the impact of imbalances in the scale of network parameters during the learning process.

The probability distribution for each pixel across different classes is determined by the output activation function in the neural network. In our case, Softmax is used for the output layer. During training, the weights of the model are adjusted based on the output in order to minimize the loss function, which compares the predicted probabilities with the actual values.

There is a class imbalance in the dataset (Table 1), which affects the quality of model learning. In the article [35], it is proposed to use two hybrid loss functions, including contributions of various losses, combined cross-entropy and Dice losses.

The cross-entropy loss for a multi-class task as:

$$CE = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=0}^{K-1} q_{i,k} \log p_{i,k}, \tag{4}$$

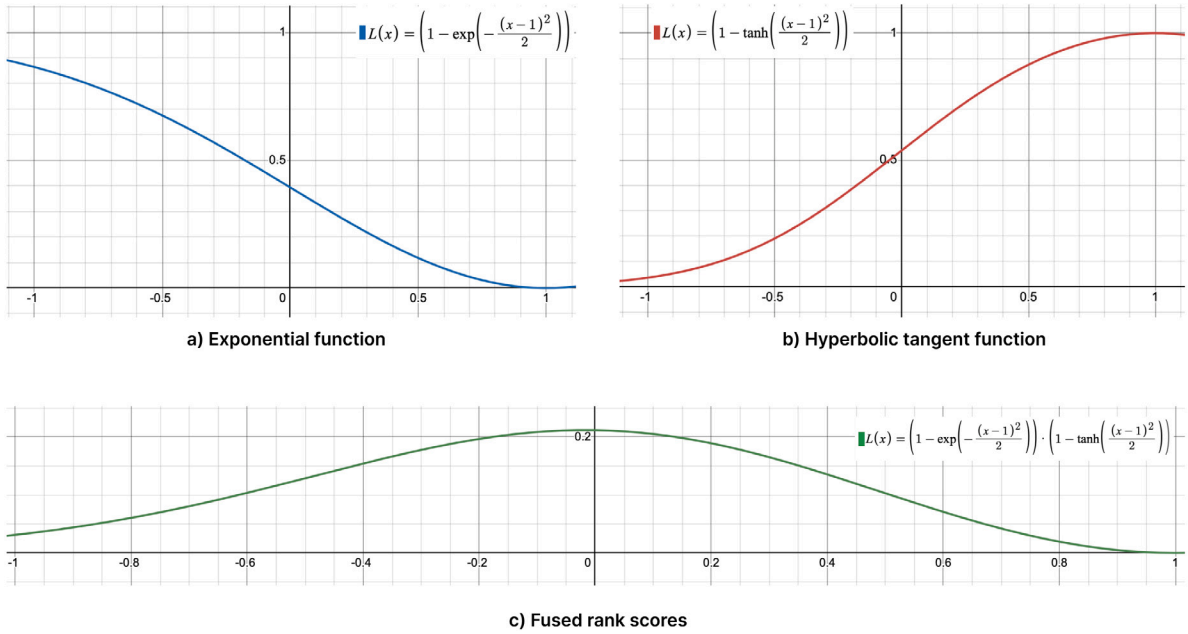

Fig. 3. Graphs of fuzzy ranks based on nonlinear functions denoting the probability of a class of sample data.

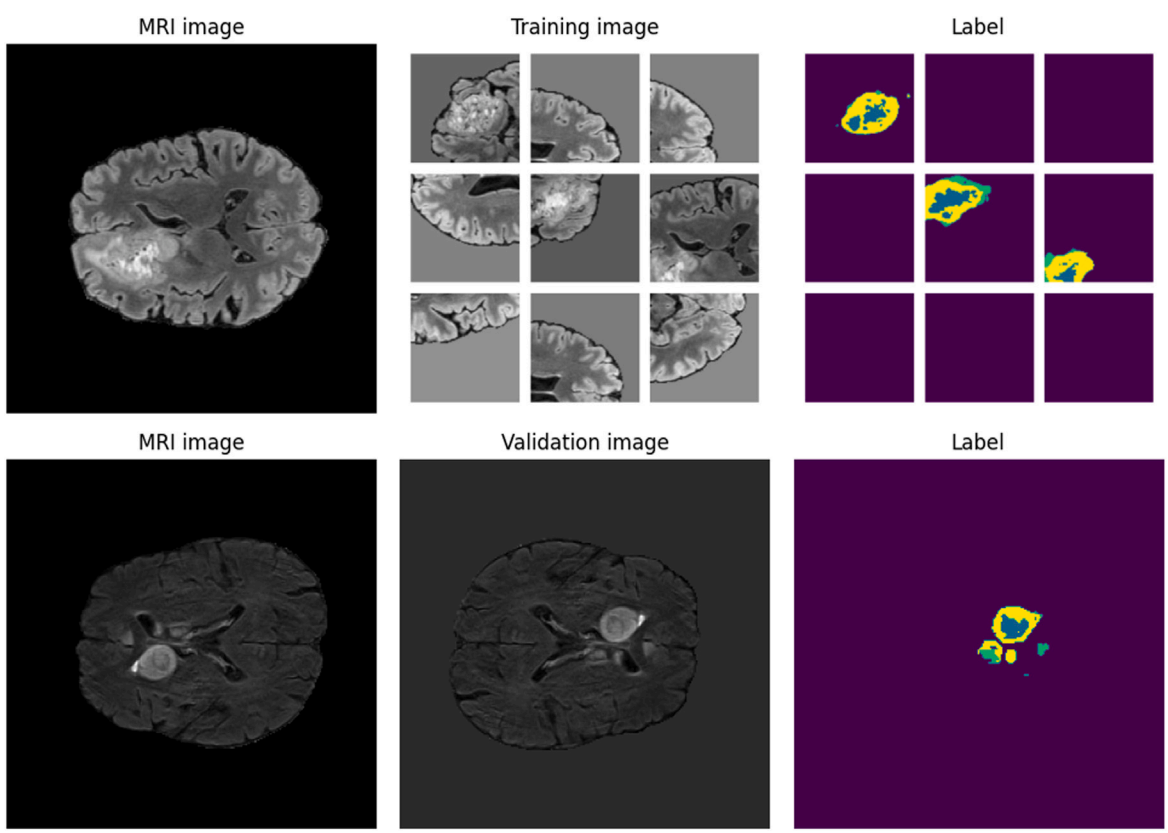

Fig. 4. Visualization of the preprocessing procedures (a single slice from 3d image). A source image in the left column, a preprocessed (input) image in the central column, preprocessed label in the right column. Class ET — yellow, WT — green, TC — blue.

where $q_{i,k} \in 0, 1$ and $p_{i,k} \in 0, 1$ denote the manual annotation and the predictions of the Softmax at each pixel i for class k respectively, and N represents the number of pixels.

The Dice loss for each subclass can be expressed as:

$$DiceLoss = 1 - \frac{2\sum_{i=1}^{N} p_{i,k} q_{i,k}}{\sum_{i=1}^{N} p_{i,k} + \sum_{i=1}^{N} q_{i,k}}, \tag{5}$$

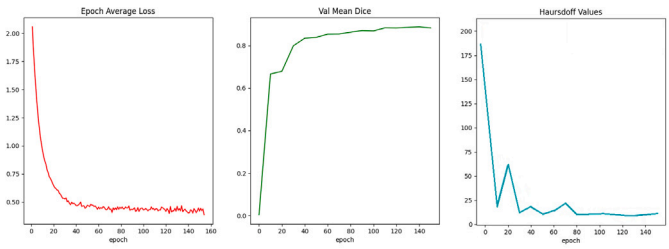

Fig. 5. Loss and learning curves for the SegResNet model.

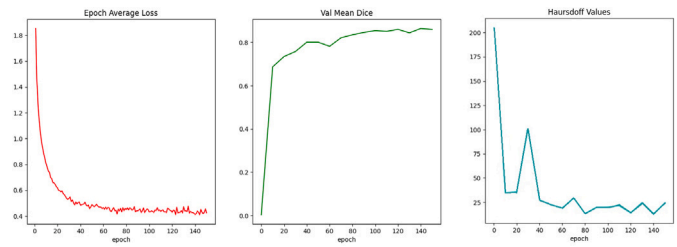

Fig. 6. Loss and learning curves for the UNETR model.

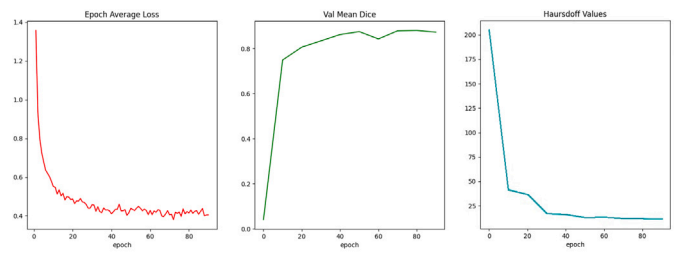

Fig. 7. Loss and learning curves for the SwinUNETR model.

The Dice Loss function is used as a loss function in combination with Cross-Entropy, which is called DiceCE Loss, to overcome the effect of class imbalance between tumor/edema and healthy tissues. This function returns the weighted sum of two losses.

2.4. Quality estimation

There are various methods for assessing the accuracy of segmenting pathological brain tissue using different performance indicators. After reviewing a number of related studies [36,37], researchers identified the Dice coefficient as a priority [38]. Furthermore, some experts have calculated Hausdorff distances [39], which are also considered important metrics for evaluating the effectiveness of a model's tumor segmentation. Therefore, the quality assessment of the segmentation of the base models as well as the proposed ensemble is performed using Dice [40] and Hausdorff [41] metrics. This choice is based on the review of the indicators [42].

The Dice metric is well-known in segmentation tasks and evaluated according to the following equation.

$$DSC = \frac{2TP}{2TP + FP + FN}, \tag{6}$$

where TP is a true positive result; FN is a false negative result; FP is a false positive result. The coefficient measures the overlap of two samples, the indicator varies from 0 to 1, where 1 means complete overlap.

The Hausdorff metric is based on spatial distance and is used to estimate the overall segmentation accuracy. The distance between two sets of endpoints A and B is determined by the following equation:

$$HD(A, B) = \max(h(A, B), h(B, A)), \tag{7}$$

where $h(A, B)$ is called the directed Hausdorff distance and is given by

$$h(A, B) = \max_{a \in A} \min_{b \in B} \|a - d\|, \tag{8}$$

where $\|a - b\|$ is some norm, e.g. Euclidean distance.

3. Experimentation and results

3.1. Results of base models

The method of segmentation of 3D medical images has been implemented in Python, and the choice is due to the convenience in implementing many deep learning algorithms. The program for learning validation and visualization of results is based on the Pytorch Lightning and MONAI [43] frameworks.

The SegResNet network uses a ResBlock and employs skip connections and two convolution blocks, but it does not include the variational autoencoder. The activation function and argument are rectified linear unit (ReLU), the number of output channels for the initial convolution level is 16 and the relative probability of zeroing the element is 0.2. The UNETR model uses convolutional and residual blocks. The type of object normalization is based on instances. The UNETR model uses 48 dimension of network feature size and uses gradient checkpointing for reduced memory usage.

The size of the base models SwinUNETR, UNETR, and SegResNet is 754.03 MB, 1.23 GB, and 56.53 MB, respectively. The learning process averaged 150 epochs. Neural network training is performed using a remote computational environment with the following hardware configuration: Intel Skylake 4 core CPU, 2 Nvidia Tesla T4 GPUs, and 29 Gigabytes of RAM. The training time was 10 h.

Training time dice loss, as well as training time validation dice and Haursdoff distance metric curves are shown in Figs. 5, 6, and 7 for the SegResNet, UNETR and SwinUNETR models, respectively.

The saturation of the curves on the right-hand side of all the graphs allows us to determine whether the models have been trained successfully. Additionally, the maximum level of performance for all the models is achieved after approximately 80 epochs, with none of the models showing signs of overfitting during training.

3.2. Results of the ensemble model

The implementation of an ensemble based on fuzzy ranking begins with the preparation of the input data. Using an MRI scan of the brain, three pre-trained models make predictions for each class (background, WT, ET, TC). The 3D arrays obtained at the output of these models are converted into three 1D arrays, each measuring 8 928 000 voxels and depleted into a single 3 × matrix 8 928 000 voxels for convenience of calculations. As a result, four 2D arrays (corresponding to 4 classes of tumors) are obtained, which are fed to the ensemble, and further calculations are made.

This implementation helps to calculate all mathematical steps described in the 2.2.2 chapter faster. As a result of the ensemble's work, the final segmentation of glioblastoma is obtained, taking into account the characteristic features of all its constituent models, which makes it possible to improve the quality of prediction relative to the basic segmentation models. Fig. 8 shows a comparison of the best and worst segmentation results of a brain tumor between individual models and the ensemble. The axial slice 80 of the BraTS-GLI-00525-001 image layer from the test set is chosen as the worst case example, and the BraTS-GLI-00479-001 image is the best example.

In the final segmentation, we have compared with the results of individual models, and due to the clear ranking, all features of the predictions are taken into account, thereby reducing the number of outliers

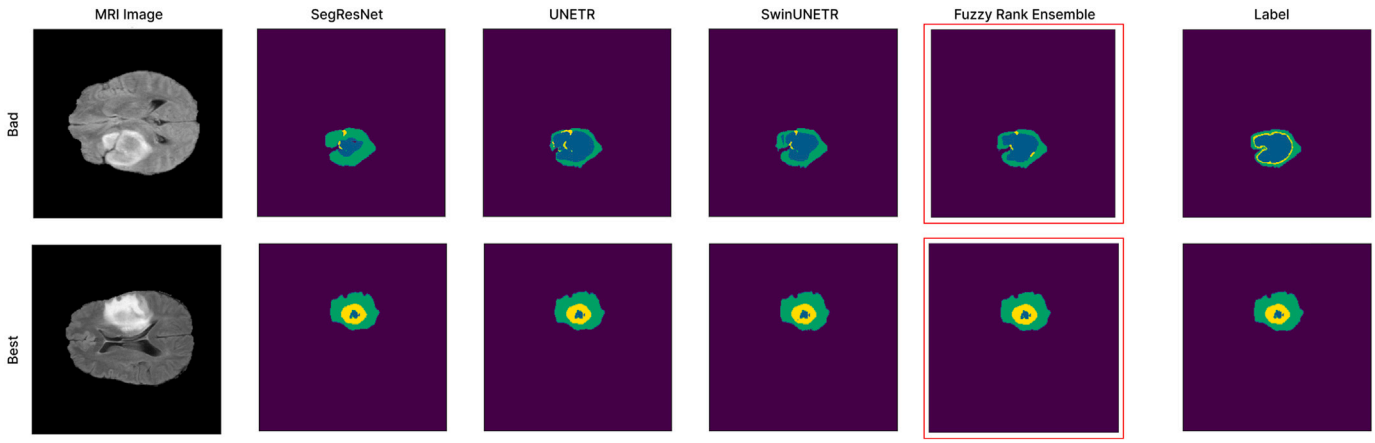

Fig. 8. Visualization of predictions. From left to right: Input MRI image, prediction of SegResNet model, prediction of UNETR model, prediction of SwinUNETR model, result of fusion using proposed method, ground truth label.Class ET — yellow, WT — green, TC — blue.

Table 2
Performance comparison in terms of Dice score, and estimation of statistical significance of the difference between a base model and ensemble in terms of *p*-value calculated based on the Wilcoxon signed-rank test.

Class	Model	Mean ± Std	Median (IQR)	*p*-value
WT	SegResNet	0.821 ± 0.157	0.889 (0.151)	**0.000**
	SwinUNETR	0.827 ± 0.150	0.877 (0.140)	**0.000**
	UNETR	0.780 ± 0.192	0.864 (0.274)	**0.000**
	Ensemble	**0.832 ± 0.153**	**0.898 (0.170)**	–
ET	SegResNet	0.879 ± 0.100	0.909 (0.072)	0.494
	SwinUNETR	0.851 ± 0.135	0.895 (0.095)	**0.000**
	UNETR	0.863 ± 0.236	0.899 (0.086)	**0.000**
	Ensemble	**0.880 ± 0.095**	**0.910 (0.079)**	–
TC	SegResNet	0.848 ± 0.154	0.904 (0.110)	**0.007**
	SwinUNETR	0.825 ± 0.175	0.885 (0.128)	**0.000**
	UNETR	0.778 ± 0.218	0.865 (0.176)	**0.000**
	Ensemble	**0.851 ± 0.156**	**0.907 (0.118)**	–

Table 3
Comparison of performance in terms of Hausdorff distance, and estimation of statistical significance of the difference between a base model and ensemble in terms of *p*-value calculated based on the Wilcoxon signed-rank test.

Class	Model	Mean ± Std, mm	Median (IQR), mm	*p*-value
WT	SegResNet	39.045 ± 28.079	39.064 (53.578)	**0.002**
	SwinUNETR	38.531 ± 26.962	**34.234 (1.817)**	**0.003**
	UNETR	54.331 ± 28.085	59.313 (47.799)	**0.035**
	Ensemble	**38.406 ± 28.575**	37.603 (54.490)	–
ET	SegResNet	17.156 ± 21.135	8.124 (13.594)	**0.003**
	SwinUNETR	**13.410 ± 15.941**	7.681 (8.108)	**0.003**
	UNETR	35.780 ± 31.337	31.701 (51.475)	**0.030**
	Ensemble	15.962 ± 21.185	**7.211 (10.310)**	–
TC	SegResNet	**10.777 ± 10.877**	8.062 (6.705)	**0.002**
	SwinUNETR	11.040 ± 9.467	8.307 (5.480)	**0.002**
	UNETR	41.603 ± 33.426	31.890 (60.727)	**0.020**
	Ensemble	12.061 ± 12.411	**8.062 (4.657)**	–

in the final image. The example of bad segmentation in Fig. 8 illustrates how each model evaluates the prediction of a class differently, while the ensemble corrects their inaccuracies

The statistical analysis of the ensemble with comparison on the base neural network is performed on an unseen test subset. In Tables 6–9 expanded descriptive statistics are presented that summarize the central tendency, dispersion, and shape of a dataset's distribution, excluding NaN values for all three base models and for the fuzzy rank ensemble test time performance. Tables 6–9 show the calculation of two metrics to analyze the operation of three models and the proposed method to determine the pathological glioblastoma cells WT, TC, ET and background.

The Wilcoxon signed rank test has been utilized to estimate a significant difference between the base models and the ensemble. The result of this comparison is presented in Table 2. As we can see, the performance of the ensemble is better than that of the base models in terms of Dice scores for all classes. However, some base models show an almost similar performance to the ensemble's performance. For example, the ensemble shows the mean Dice score of 0.880 ± 0.0953 and the median quality of 0.910 (0.079) for ET class segmentation, while SegResNet shows the mean Dice score 0.879 ± 0.100 and the median quality 0.909 (0.072).

For the WT class, SwinUNETR has the best result among the basic models with a mean dice score of 0.827 ± 0.150 and a median quality of 0.877 (0.140). For the TC class, the best SegResNet base model has a mean dice score 0.848 ± 0.154 and the median quality of 0.904 (0.110). Consequently, due to comparable results for internal areas, the segmentation of the external border of the tumor improved when using the ensemble.

On the other hand, in terms of Hausdorff distance, as presented in Table 3 proposed ensemble does not show a significant difference with the best base networks. Namely, without significant difference the results of the ensemble is equivalent to SegResNet and SwinUNETR.

In Fig. 9, we can see the distribution of Dice and Hausdorff metrics based on a test sample to compare three trained samples and the proposed ensemble with respect to the prediction of each tumor class. The largest data spread of the model is shown in the enhancing tumor class; however, an ensemble based on fuzzy logic allows us to reduce the variability of the data.

Table 4 compares the proposed method with some other published glioblastoma segmentation methods based on the BraTS2020 dataset. Snehal Rajput et al. [44] proposed the CCSAv1 triplanar ensemble, which is an ensemble of axial, sagittal, and coronal models with the CCSAv1 attention mechanism. Yuan Z. et al. [45] proposed a hierarchical disconnected convolutional network (HDC-Net), where 3D convolution with a novel hierarchical decoupled convolution (HDC) module that can explore multiscale representations in multiple spatial contexts with high efficiency. Ali et al. [46] proposed an ensemble of a set of 2D and 3D models to better exploit their respective advantages. Experimental results on the BraTS 2020 test dataset showed that the dice score was 0.79 for ET, 0.87 for WT and 0.83 TC.

On the basis of the obtained results, it can be concluded that an indistinctly ranked ensemble, on average, works more accurately than separately applied individual models. It also competes in comparison with other proposed ensembles that were published earlier. Therefore, the method we have developed improves the quality of the overall prediction ability of the proposed model compared to its concurrent models.

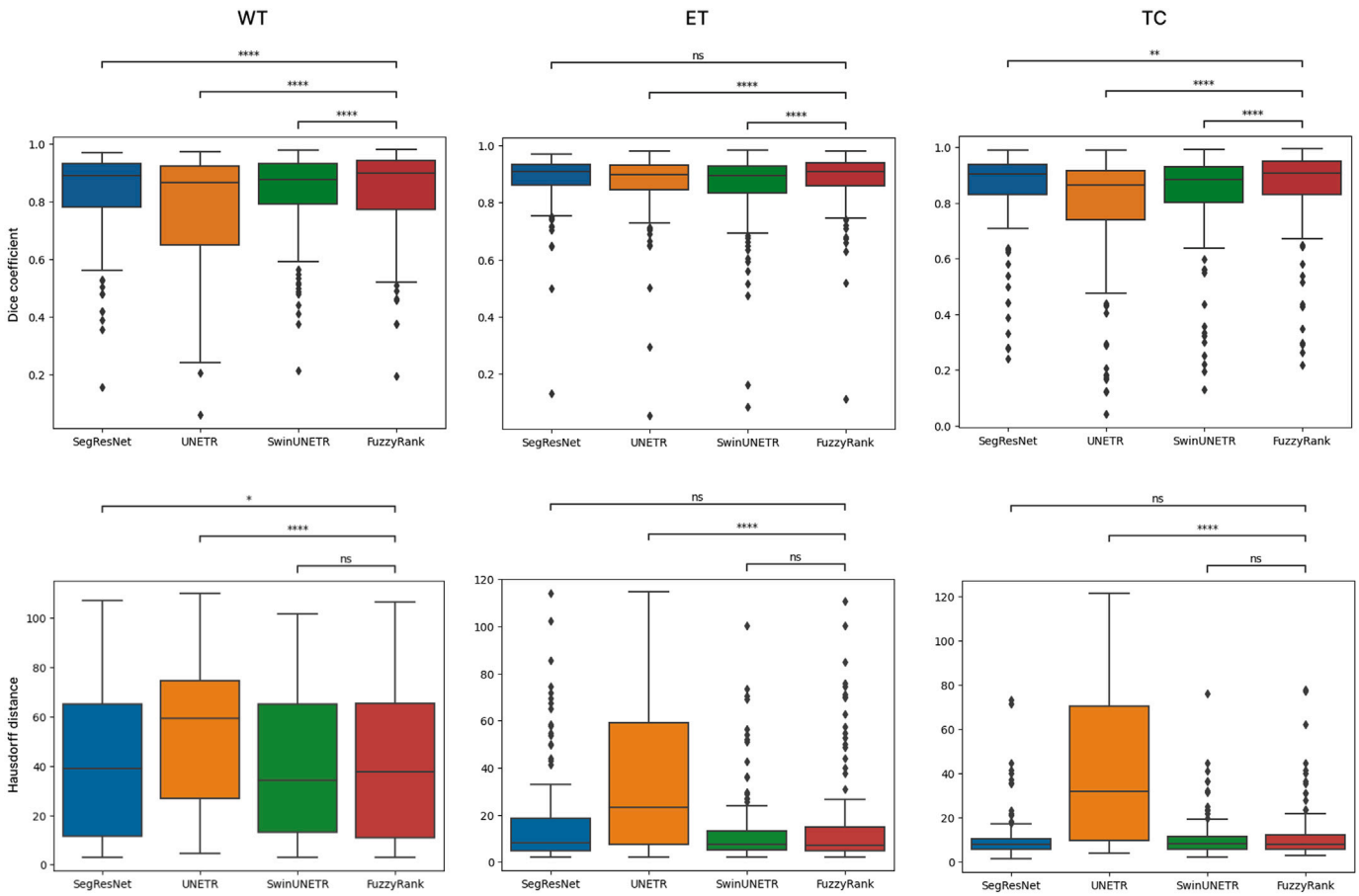

Fig. 9. The boxplots of ablation models in terms of Dice and Hausdorff metrics. The Wilcoxon signed rank test p-value annotation legend:ns: $5\times10^{-2} < p \leq 1$, *: $10^{-2} < p \leq 5\times10^{-2}$, **: $10^{-3} < p \leq 10^{-2}$, ***: $10^{-4} < p \leq 10^{-3}$, ****: $p \leq 10^{-4}$.

Table 4
Comparison of the proposed model with state-of-the-art methods.

	Mean dice score				Mean Hausdorff distance, mm			
	WT	ET	TC	Average	WT	ET	TC	Average
SegResNet [27]	0.821	0.879	0.848	0.850	39.045	17.156	10.777	22.326
UNETR [28]	0.780	0.863	0.778	0.807	54.331	35.780	41.603	43.904
SwinUNETR [32]	0.827	0.851	0.825	0.834	38.531	13.410	11.040	20.994
Ensemble CCSAv1 [44]	0.863	0.699	0.777	0.779	**7.173**	35.713	14.529	19.138
Ensemble of 2D/3D [46]	**0.871**	0.748	0.748	0.789	9.428	**3.929**	**10.090**	**7.815**
HDC-Net [45]	0.845	0.677	0.735	0.759	34.21	53.26	33.56	40.35
Proposed ensemble	0.832	**0.880**	**0.851**	**0.855**	38.406	15.962	12.061	22.143

4. Discussion and conclusion

Segmentation of a brain tumor image is crucial to accelerate the diagnosis and treatment of a patient, as well as the further development of the use of computer vision in the field of medicine. In this article, we have developed an ensemble model to improve the segmentation of brain tumor boundaries, namely such pathological cells as the enhancing tumor, on-enhancing tumor core, and vasogenic edema.

The main novelty of this proposed idea is the application and evaluation of fuzzy ensembles [26] for segmentation tasks. We take into account all the information obtained from basic models by evaluating two nonlinear functions: an exponentially decreasing function that considers the deviation from the target for the class and a tangentially increasing function that determines the reward for a high probability of belonging to the class. Exponential and hyperbolic functions are necessary to determine the score that should be assigned to each model for the final assessment of whether a pixel belongs to a certain class (related to segmentation).

The final prediction is a combination of all the quantitative values from three neural networks, allowing us to solve the given problem more effectively. Three U-Net-based architectures are selected, namely SegResNet, UNETR, and SwinUNETR, and the model is evaluated using the BRATS 2023 MR imaging suite. As a result, the choice of class is based on the probability of prediction of each model,

As shown in the results, the proposed method is competitive compared to other recent variants of brain tumor segmentation. All methods show better efficiency in WT segmentation than in TC and ET segmentation in terms of Dice metrics. This is due to the fact that the task of segmenting large objects can easily obtain a higher metric score than small objects. However, it can be noted that such a volume displacement does not affect the Hausdorff metric. As a result, using the ensemble, it is possible to improve the quality of segmentation of the boundaries of a brain tumor on an MRI scans.

Table 5
Estimation of the model's stability to various image transformations: Gaussian noise – standard deviation of distribution; Space spike noise – value for the log-intensity of the k-space version of the image; Adjust contrast – value for the log-intensity of the k-space version of the image; Bias field – coefficient of the displacement field.

	Value	Median Dice score (*p*-value)			Median Hausdorff distance, mm (*p*-value)		
		WT	ET	TC	WT	ET	TC
Baseline	0.0	0.898	0.910	0.907	37.603	7.211	8.062
Gaussian noise	0.05	0.326 (0.000)	0.192 (0.000)	0.500 (0.000)	103.344 (0.000)	88.391 (0.000)	100.729 (0.000)
	1.0	0.325 (0.000)	0.192 (0.000)	0.500 (0.000)	103.344 (0.000)	88.391 (0.000)	100.729 (0.000)
	20.0	0.324 (0.000)	0.191 (0.000)	0.494 (0.000)	103.846 (0.000)	88.289 (0.000)	100.600 (0.000)
Spike noise	5.0	0.163 (0.000)	0.280 (0.000)	0.154 (0.000)	89.370 (0.000)	102.071 (0.000)	105.085 (0.000)
	15.0	0.162 (0.000)	0.280 (0.000)	0.153 (0.000)	89.381 (0.000)	102.079 (0.000)	105.088 (0.000)
	17.0	0.162 (0.000)	0.279 (0.000)	0.153 (0.000)	89.385 (0.000)	102.081 (0.000)	105.094 (0.000)
Adjust contrast	0.1	0.182 (0.000)	0.404 (0.000)	0.069 (0.000)	100.264 (0.000)	183.882 (0.000)	102.369 (0.000)
	0.3	0.324 (0.000)	0.590 (0.000)	0.269 (0.000)	98.030 (0.000)	177.045 (0.000)	104.54 (0.000)
	0.5	0.447 (0.000)	0.776 (0.000)	0.556 (0.000)	89,785 (0.000)	101,770 (0.000)	95,257 (0.000)
	0.7	0.606 (0.000)	0.876 (0.000)	0.802 (0.000)	82.325 (0.000)	86.519 (0.000)	91.175 (0.000)
	0.9	0.831 (0.000)	0.895 (0.308)	0.887 (0.086)	69.079 (0.000)	26.863 (0.000)	11.402 (0.000)
Bias field	0.1	0.877 (0.000)	0.903 (0.000)	0.884 (0.000)	46.303 (0.000)	9.274 (0.000)	9.220 (0.021)
	0.2	0.749 (0.000)	0.858 (0.000)	0.851 (0.000)	66.317 (0.000)	24.431 (0.000)	10.66 (0.000)
	0.4	0.436 (0.000)	0.666 (0.000)	0.690 (0.000)	74.384 (0.000)	74.672 (0.000)	75.838 (0.000)
	0.6	0.284 (0.000)	0.352 (0.000)	0.320 (0.000)	75.597 (0.000)	80.551 (0.000)	81.749 (0.000)
	0.8	0.174 (0.000)	0.165 (0.000)	0.083 (0.000)	77.365 (0.000)	82.386 (0.000)	83.853 (0.000)
	1.1	0.088 (0.000)	0.011 (0.000)	0.001 (0.000)	77.926 (0.000)	84.794 (0.000)	86.489 (0.000)
Crop	–	0.886 (0.440)	0.899 (0.056)	0.865 (0.064)	14.177 (0.000)	7.021 (0.200)	7.348 (0.097)

Table 6
SegResNet test time performance.

	Dice score			Hausdorff distance, mm		
	TC	WT	ET	TC	WT	ET
mean	0.848	0.822	0.881	10.777	39.045	17.156
std	0.154	0.157	0.101	10.878	28.080	21.135
min	0.239	0.155	0.129	1.414	2.828	2.236
25%	0.829	0.779	0.863	5.831	11.576	4.899
50%	0.904	0.889	0.909	8.062	39.064	8.124
75%	0.939	0.930	0.935	10.488	65.154	18.493
max	0.988	0.969	0.970	73.273	107.023	114.000

Table 7
SwinUNETR test time performance.

	Dice score			Hausdorff distance, mm		
	TC	WT	ET	TC	WT	ET
mean	0.826	0.827	0.852	11.041	38.532	13.410
std	0.175	0.150	0.135	9.467	26.963	15.941
min	0.128	0.214	0.084	2.236	2.828	2.000
25%	0.802	0.791	0.835	6.000	13.191	5.196
50%	0.885	0.877	0.895	8.307	34.234	7.681
75%	0.930	0.931	0.930	11.489	65.008	13.304
max	0.991	0.977	0.983	76.269	101.573	100.444

The quality indicators of the ensemble developed here with individual base architectures and other published methods are analyzed. Based on this analysis, it can be said that the ensemble based on fuzzy fusion significantly improves the quality of individual models so that the average accuracy of brain tumor segmentation becomes 0.885 ± 0.134, and according to the Hausdorff distance 22.143 ± 9.531. The statistical significance of the differences between the models and the ensemble is confirmed using the Wilcoxon signed rank test, and the *p*-value is below 0.005.

Analyzing the quality indicators achieved by the basic architectures, it can be seen that the SegResNet model copes better with the task, both on average (Table 4) and for individual ET and TC classes (Table 2). This can be explained by the fact that the architecture presented is based on the regulation of the auto-encoder for fast calculations. The SwinUNETR architecture shows the best result for segmentation of the WT class, and UNETR lags behind the indicators in this test.

The outliers in Fig. 9 can be explained by the fact that the segmentation area of interest is small and difficult to cover. The imbalance of data is observed not only between the brain tumor and the background but also between the subdomains of the tumor. At the same time, the tumor and especially its subdomains have complex boundaries and structure, which is due to the penetration of malignant tissues into healthy ones.

Finally, we discuss the robustness of the proposed model to image disturbances. For this estimation, we analyzed the following disturbances: additive Gaussian noise, spike noise, random contrast adjustment, bias field [47]. All mentioned disturbances have been applied to a source image.

The Table 5 shows a comparison of changes in the median values of the segmentation quality metrics for the proposed ensemble, depending on the level of Gaussian noise, spike noise, random contrast adjustment, and bias field in the test sample. The Wilcoxon test results are shown in parentheses in comparison to images without noise.

In the experiment with adjusting the contrast using gamma conversion, we used both contrast reduction (a coefficient less than 0) and enhancement (a coefficient greater than 0). Based on the results, we found that a 20% change in contrast did not significantly affect the quality of the segmentation. However, in the case of the bias field, even a small change in coefficient (0.1) led to a decrease in segmentation accuracy. This suggests that low-frequency signals, which affect pixel brightness, can have a significant impact on model predictions.

In an experiment with the addition of Gaussian noise and spatial noise [48], we can see a significant performance decrease, which may be explained by the peculiarities of the data preprocessing step. Namely, for the normalization only nonzero values are used, which exclude an image background from statistics, but it is not true for noised images, where a background has a significant effect on image density distribution, and, as a consequence, leading to differences in the normalization.

As a result of our research, we found that moderate multiplicative changes in pixel intensity had little effect on the quality of segmentation. However, the noise in the image could affect the predictions made by the model. This factor can be addressed by further training the models or additional preprocessing of noised data. However, this flaw cannot be considered significant due to the typical absence of such noise in the raw MRI data used for this task.

Table 8
UNETR test time performance.

	Dice score			Hausdorff distance		
	TC	WT	ET	TC	WT	ET
mean	0.778	0.781	0.863	41.603	54.331	35.780
std	0.218	0.193	0.120	33.426	28.086	31.337
min	0.040	0.058	0.051	4.000	4.583	2.236
25%	0.739	0.650	0.845	9.849	26.833	7.483
50%	0.865	0.864	0.899	31.890	59.313	23.388
75%	0.915	0.924	0.931	70.576	74.632	58.958
max	0.989	0.972	0.981	121.417	109.813	114.586

Table 9
Fuzzy rank ensemble test time performance.

	Dice score			Hausdorff distance		
	TC	WT	ET	TC	WT	ET
mean	0.851	0.832	0.880	12.061	38.407	15.962
std	0.157	0.153	0.105	12.411	28.575491	21.185
min	0.217	0.193	0.111	3.162	3.000	2.236
25%	0.830	0.771	0.860	5.745	11.000	4.690
50%	0.907	0.898	0.910	8.062	37.603	7.211
75%	0.948	0.941	0.939	12.450	65.490	15.000
max	0.993	0.979	0.982	77.929	106.405	110.684

Despite the fact that the ensemble model shows an improvement in the segmentation results, the quality gain is not as significant as in the classification task reported in [26]. This can be explained by the fact that segmentation task is more difficult to resolve. This can lead to both improved results for one metric (Dice) and worse results for another (Hausdorff distance).

Next, despite the fact that boundary estimation is the main goal of medical examination, the Hausdorff distance metric is based on the maximum distance evaluation, which makes it difficult to direct usage as a loss function. Meanwhile, the HD loss function proposed in [49] significantly increases the computational cost of the training process. Thus, considering that the main goal of the research is the extension of the fuzzy ensemble, the more cost-effective and conventional loss function was selected. In the future, it makes sense to utilize HD loss with or without fusion with Dice or Twersky loss functions, and contextual information about the pixel location in the ensemble may be considered to obtain better results.

CRediT authorship contribution statement

Daria Valenkova: Writing – original draft, Visualization, Software, Investigation. **Asya Lyanova:** Writing – original draft, Visualization, Validation, Investigation, Conceptualization. **Aleksandr Sinitca:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Ram Sarkar:** Writing – review & editing, Methodology, Formal analysis, Data curation, Conceptualization. **Dmitrii Kaplun:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The training pipelines as well as used datasets are available in Kaggle platform:

1. SegResNet training pipeline [50].
2. UNETR training pipeline [51].
3. SwinUNETR training pipeline [52].
4. Ensemble analysis pipeline [53].

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